

Lab 2 – Harshad Numbers

Requirements

Prompts the user for the name of the input and output files (the user enters a single filename, “filename” where the input is “filename.in” and the output is “filename.out”), the input file contains a list of numbers (one on each line), the program determines if the number is a Harshad number or not and prints the number and that determination to the output file (one on each line).

A Harshad number is one which is evenly divisible by the sum of the digits of the number. For example, the number 18 has digits 1 and 8, $1+8=9$ and 18 is divisible by 9 therefore 18 is a Harshad number. The number 28 has digits 2 and 8, $2+8=10$ and 28 is not divisible by 10 therefore 28 is not a Harshad number.

Your program must implement two functions that are contained within a module (.h and .cpp file pair), the functions are highlighted in the C version of `main()` below; the parameters and return values of your C++ functions `GetFilename()` and `IsHarshad()` must match what is shown below. Note that your program must use a C++ version of `main()` which you must write and not the pseudo-code version given below. Also, your `GetFilename()` function should verify that the base filename entered is valid by ensuring that a file “filename.in” exists and is readable. The user should be prompted again if the filename entered is not valid until a valid name is entered.

The contents of your input file should be such that there is a single number on each line of the file, your program will check each number in order. Be sure to use a TEXT editor to put the values in the input file (DO NOT USE MS-WORD).

```
/* Get maximum number */
Basename = GetFilename();

/* Loop through reading all the lines of the file "Basename.in" */
/* Check to see if the number is an Harshad number or not */
if (IsHarshad(CandidateNumber) == 1)
{
    /* Indicate a Harshard number in the output file "Basename.out"*/
} /* if() */
else
{
    /* Indicate NOT a Harshard number in the output file "Basename.out" */
} /* if() */
} /* loop() */
```

Assessment Criteria

- 6pts – Complete and accurate implementation of the design specifications
- 2pts – Hierarchical module files

- 2pts – Proper comments and code structure

Submission Requirements

- Submit only your source code (i.e., main C++ source code file, module C++ source code file and module header file) – do not submit the entire Visual Studio project
- A document containing screenshots of your program with different cases and the contents of your input and output files to demonstrate full functionality (PDR preferred)

Note that I will be compiling and running your program and testing it with my data file, the maximum grade is 1pt if I cannot compile and run your program from the files that you have submitted.

Objective

The purpose of this lab is to introduce C++ code, hierarchical (module-based) source code partitioning, and file input/output.